

Northeastern University

**Project Report – *Image Captioning with CLIP
and GPT-2***

(Generative Project Milestone - 2)

IE 7615 - Deep Learning for AI

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Group 3

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1. Architecture — CLIP + GPT-2 Integration

We integrated the CLIP image encoder with the GPT-2 language decoder using **prefix conditioning**. This approach injects visual information by prepending learned pseudo-tokens to GPT-2's input sequence.

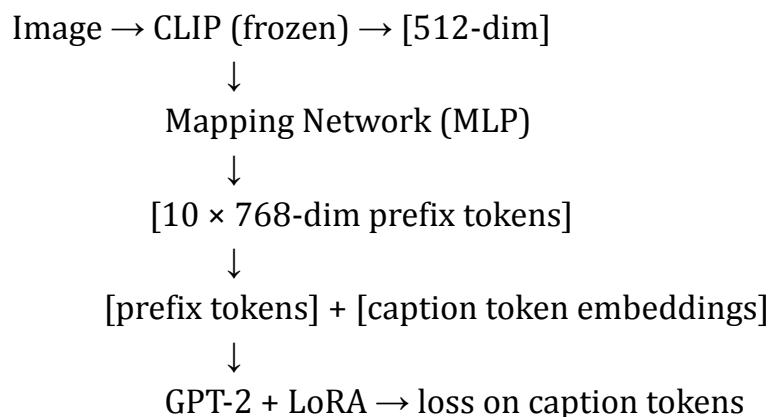
Three components:

CLIP ViT-B/32 (Frozen) — Encodes each image into a 512-dimensional embedding. CLIP is kept frozen and used only for pre-computing embeddings before training begins, freeing GPU memory during the training loop.

Mapping Network (Trained from scratch) — A 3-layer MLP that projects CLIP's 512-dim vector into a sequence of 10 prefix tokens in GPT-2's 768-dimensional embedding space. Architecture: $\text{Linear}(512 \rightarrow 512) \rightarrow \text{GELU} \rightarrow \text{LayerNorm} \rightarrow \text{Linear}(512 \rightarrow 512) \rightarrow \text{GELU} \rightarrow \text{LayerNorm} \rightarrow \text{Linear}(512 \rightarrow 7680)$, reshaped to 10×768 .

GPT-2 with LoRA (Fine-tuned) — GPT-2 (124M params) generates captions conditioned on the prefix tokens. We apply LoRA (rank=8, alpha=16) to the attention layers (c_attn, c_proj), reducing trainable parameters to approximately 0.5–1% of the full model.

Embedding injection method: The mapping network's output is concatenated with GPT-2's token embeddings along the sequence dimension. An extended attention mask ensures GPT-2 attends to both the visual prefix and text tokens. Loss is computed only on caption token positions, not on the prefix.



2. Training Configuration

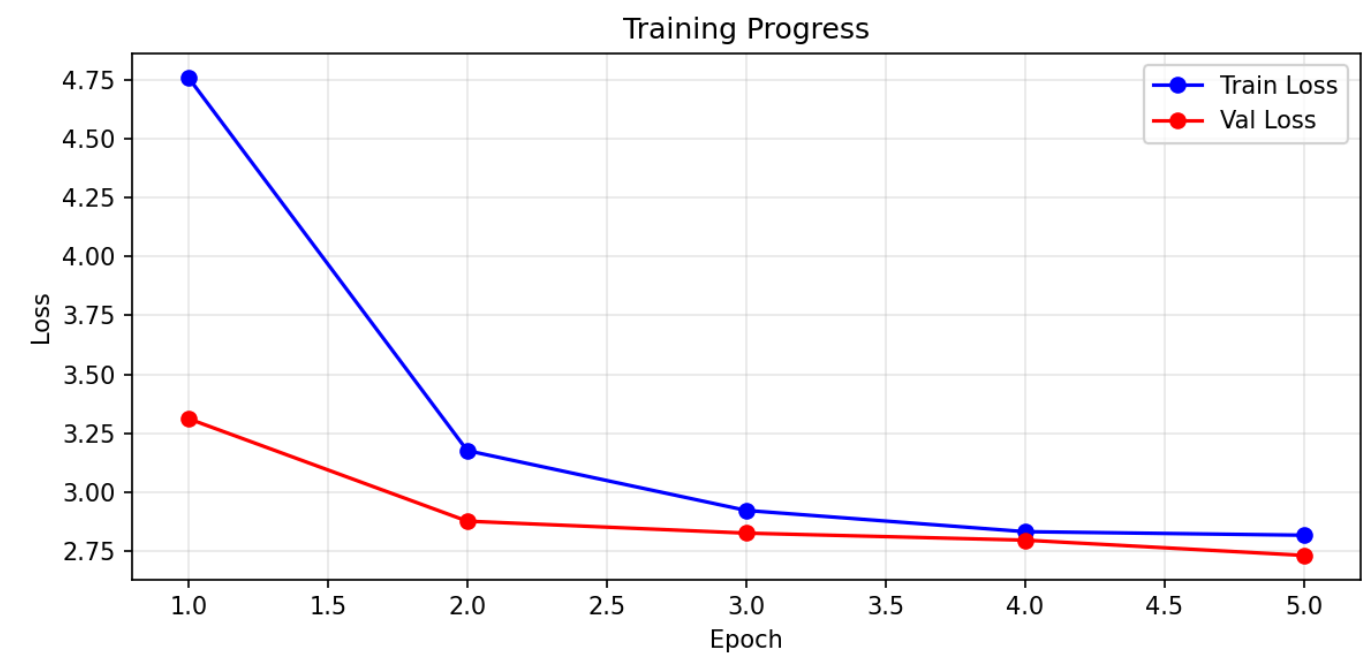
Parameter	Value
Training pairs	2,500 (COCO Karpathy train split)
Validation pairs	500 (COCO Karpathy validation split)
Epochs	5
Batch size	16 (effective: 32 with gradient accumulation)
Learning rate	2e-4 with linear warmup (100 steps) and decay
Optimizer	AdamW (weight decay: 0.01)
Gradient clipping	Max norm 1.0
Max caption length	64 tokens
Prefix length	10 tokens
LoRA rank / alpha	8 / 16
Hardware	Google Colab T4 GPU

3. Training Log

Epoch	Train Loss	Val Loss	Observation
1	4.7619	3.3115	Model outputs incoherent text with heavy repetition
2	3.1756	2.8757	Recognizes food-related content, still very generic
3	2.9212	2.8248	Captions become more structured, repetition persists
4	2.8310	2.7950	Starts identifying scene elements (plates, tables, glass)
5	2.8159	2.7300	Best val loss; nucleus sampling produces coherent captions

Key observations from the training log:

The largest improvement occurred between epochs 1 and 2, where train loss dropped by 1.59 points (33% reduction). After epoch 2, the model entered a phase of gradual refinement with diminishing returns per epoch. Validation loss decreased consistently across all 5 epochs without diverging from train loss, indicating no overfitting at this scale.



4. Early Caption Samples

Ground truth (GT) : "a table in the sun with a white cloth, beef sandwich, glass of beer, and bottle of wine."

Epoch	Greedy	Nucleus
1	"a large plate of chicken and cheese. I'm not sure if I'm going to eat it..." (degenerates into loops)	"my food with vegetables and bread. When I make the croissants and bread..." (off-topic but structured)
3	"a plate of pizza with a pizza on top. The pizza is on top of a plate..." (repetitive but food-related)	"a big plate of pizza with a large ice cream cone..." (creative, partially relevant)
5	"a plate of food is sitting on top of a plate. The food is covered in a sauce..." (recognizes food/plate)	"a large pizza on a plate and a brown paper towel next to it. The pizza is decorated with toppings..." (most coherent)

The model progressed from incoherent text (epoch 1) to recognizing food scenes and tableware (epoch 5), though it still defaults to generic food descriptions rather than identifying specific items like sandwiches or beer.

5. Decoding Strategy Comparison

We compared three decoding strategies on 5 validation images using the best checkpoint (epoch 5, val_loss=2.73):

Strategy Descriptions

Greedy decoding selects the highest probability token at each step. It is deterministic and fast, but prone to repetition loops since it always follows the single most likely path.

Beam search (beam width=5) maintains the top 5 candidate sequences at each step and selects the overall highest-scoring complete sequence. It explores more of the output space than greedy but still tends toward high-probability generic phrases.

Nucleus sampling (top-p=0.9, temperature=0.8) samples from the smallest set of tokens whose cumulative probability exceeds 90%, with a temperature of 0.8 to slightly sharpen the distribution. This introduces controlled randomness, producing more diverse and natural-sounding captions.

Comparison Results



GT: a man working on ski bindings with other ski gear around
Greedy: a man is riding a skateboard on a snowboard. The man is
Beam: a man riding a skateboard on a snowboard. The man is we
Nucleus: a man and a woman playing a game of computer tennis. Th



GT: a highway scene with focus on an exit sign.
 Greedy: a red light on a street with a red light on the side. T
 Beam: a red light on the side of a street. A red light is on
 Nucleus: a street with two cars parked behind it. The cars are o



GT: a pair of scissors sits next to some other supplies
 Greedy: a kitchen with a large kitchen utensils and a large pot
 Beam: a couple of candles are placed on a table next to each
 Nucleus: a collection of colorful and colorful plants in a kitch



GT: a person is doing a trick over a trash can.
Greedy: a man is riding a skateboard on a sidewalk. The skatebo
Beam: a skateboarder riding a skateboard in a park. The skate
Nucleus: a person walking on the sidewalk with a skateboard. Ano



GT: a table in the sun with a white cloth, beef sandwich, g
Greedy: a plate of food is sitting on top of a plate. The food
Beam: a plate of food is sitting on top of a plate. The food
Nucleus: a person in a large plate with a bowl of food on it. Th

Summary of Observations

Strategy	Strengths	Weaknesses
Greedy	Fast, deterministic, consistent	Severe repetition loops ("The food is covered in a sauce. The food is covered in a sauce.")
Beam search (k=5)	More coherent than greedy, considers multiple paths	Still repetitive, tends toward generic high-probability phrases
Nucleus (p=0.9)	Most diverse and natural captions, best scene descriptions	Occasionally hallucinates details not present in the image

Overall finding: Nucleus sampling consistently produced the most readable and contextually relevant captions across all 5 test images. Greedy and beam search suffered from repetition that made outputs unusable beyond the first sentence. For future milestones, nucleus sampling will be our primary generation strategy, potentially combined with a repetition penalty.

6. Identified Issues & Planned Improvements

Issue 1: Repetition in greedy and beam decoding

Generated captions frequently enter loops, repeating the same phrase. This is a well-known problem in autoregressive language models.

Planned fix: Add a repetition penalty that scales down the logits of recently generated tokens during decoding. A penalty factor of 1.2–1.5 typically resolves this without hurting fluency.

Issue 2: Generic/vague captions

The model defaults to broad descriptions like "a plate of food" rather than specific content like "a beef sandwich with a glass of beer."

Planned fix: Scale training data from 2,500 to 10,000+ pairs. The current subset is too small for the model to learn fine-grained visual-semantic distinctions. We will also experiment with training for 10–15 epochs since validation loss was still decreasing at epoch 5.

Issue 3: Caption length control

Generated captions tend to be much longer than ground truth captions, especially with nucleus sampling, producing multi-sentence outputs where a single concise sentence is expected.

Planned fix: Increase the weighting of the EOS token during training, or add a length penalty during decoding to encourage shorter, more caption-like outputs.

Issue 4: Limited vocabulary grounding

The model sometimes produces words and concepts not grounded in the image content (e.g., "ice cream cone" for a food image that has no ice cream).

Planned fix: Incorporate CLIP-based re-ranking — generate multiple candidate captions and select the one whose text embedding is closest to the original image embedding in CLIP space.

7. Milestone 2 Deliverables

Deliverable	Status
CLIP image encoder integrated with GPT-2 decoder	✓
Embedding injection via prefix conditioning implemented	✓
Baseline caption generation on sample images	✓
First set of generated captions recorded	✓
Decoding strategies experimented (greedy, beam, nucleus)	✓
Training log with per-epoch metrics	✓
Early caption samples (5 images with captions)	✓
Summary of observations	✓

Team & Roles

Member	Role	Milestone 2 Contributions
Meena Periasamy	Data Pipeline & Embeddings	Pre-computed CLIP embeddings for train/val splits, parallel image download pipeline, dataset class implementation
Rithika Sankar Rajeswari	Model Architecture	Mapping network (MLP) design, prefix conditioning integration, CLIP-to-GPT-2 embedding injection
Sreevarshan Satiyamurthy	Training & Evaluation	Training loop with gradient accumulation and checkpointing, loss logging, learning rate scheduling
Sahil Mohanty	Generation & Analysis	Three decoding strategies (greedy, beam search, nucleus), comparison visualization, report and observations